

Cognition and Machine Learning

Project Proposal

Project Idea :

The primary objective of this project is to undertake the implementation of fundamental components pivotal to the realization of Artificial General Intelligence (AGI) through the utilization of machine learning methodologies. These foundational components can be categorized as *intuitive physics*, *intuitive psychology*, *causality*, *compositionality*, and *learning-to-learn*. While the application of such theoretical constructs within the domains of Artificial Intelligence (AI) and Machine Learning (ML) may initially appear unconventional and perhaps distant, historical perspectives reveal a symbiotic relationship between these fields. Notably, seminal works in AI and ML often found their genesis within the purview of Cognitive Science, Psychological Review, or Psychology journals, exemplified by the following:

Rumelhart, D., Hinton, G. & Williams, R. Learning representations by back-propagating errors , Institute for *Cognitive Science*, University of California

Ackley, D. H., Hinton, G. E., & Sejnowski, T. J. (1985). A learning algorithm for Boltzmann Machines. *Cognitive Science*, 9(1), 147–169.

Elman, Jeffrey L. (1990). Finding Structure in Time. *Cognitive Science* 14 (2):179-211.

These aforementioned concepts were researched and implemented in models, particularly within the realm of Cognitive Science research^{1, 2, 3}. The overarching aim of such endeavors can be succinctly summarized as the emulation of human perception, learning, and cognitive processes within machine systems : building machines that see, learn and think like people⁴. Hence, my approach shall transcend mere pattern recognition, emphasizing the imperative of comprehensive model construction.

Concepts :

1. Intuitive Physics:

Intuitive Physics can be conceptualized as an understanding of physical concepts like object permanence (Lake et al., 2017), enabling infants to comprehend the persistence of objects over time, even when they are not visually perceptible. This cognitive ability encompasses the recognition that objects retain their existence despite being temporarily occluded from view. Furthermore, infants also discount physically impossible trajectories (Lake et al., 2017). They

exhibit an awareness of how the behaviors or trajectories of objects conform to fundamental principles of physics, such as gravitational forces and friction. Additionally, they demonstrate a comprehension of properties like solidity (Lake et al., 2017) and analogous characteristics. Intuitive physics can be used as a tool to facilitate transfer learning in AI and ML or to create models to have a more causally proper understanding of the world. To simulate the objects' behavior and trajectories under different conditions, Battaglia et al., (2013) used Intuitive Physics Engine (IPE) which is a probabilistic simulation tool.

Physical Intuition also has a spatial and temporal facet. For example, Omniglot which is the dataset this project is going to employ was used by Li et al., (2022) to explore the spatio-temporal characterization capabilities of spiking neural networks (SNNs) by employing few-shot learning. Therefore, spatial and temporal aspects of data is a very important component of building models, especially models designed to reach to the human-level intelligence. Hence, one-shot and few-shot learning seem to be the best way for this challenge.

2. Intuitive Psychology:

Intuitive Psychology as mentioned by Lake et al., (2017) pertains to infants' capacity to attribute mental states, such as beliefs, goals, and desires, to other individuals. This cognitive ability significantly influences their learning processes and predictive reasoning. For instance, Lake et al., (2017) gives a video game example : when observing a person engage in a video game, a child can infer the intentions behind the player's actions based on their understanding of psychological states. Consequently, the child may discern certain in-game objects as detrimental, prompting them to avoid interaction, while others are perceived as advantageous, thereby motivating pursuit. This comprehension of psychological states facilitates infants' social cognition and informs their decision-making in various contexts.

Ability to differentiate between what is animate and what is inanimate may ease the process of learning. We assume that human beings are rational, and we expect them to behave in predicted ways. We know that inanimate objects are not rational and only subject to the physical rules. These concepts facilitate our understanding or modeling of the reality. As Lake et al., (2017) put forward, Model building is the hallmark of human-level learning. That is where Cognitive Science can make great contributions to the AI, by bridging the gap between machines and people.

3. Causality:

As mentioned above, both *intuitive physics* and *intuitive psychology* revolve around the construction of models. As previously noted, these elements are often lacking in conventional AI and ML techniques. Human cognition inherently involves an ongoing endeavor to construct representations of the surrounding world. This novel approach lies at the intersection of AI and Cognitive Science. Notably, as Lake et al., (2017) claimed, both intuitive physics and intuitive psychology can be seen

as causal models of reality. Lake et al., (2015) used Bayesian Program Learning (BPL) to capture the causal and compositional properties of the Omniglot dataset.

By giving image caption examples from Deep Neural Networks, Lake et al., (2017) shows that while Deep Learning models excel in pattern and object recognition within datasets, they fall short in establishing causal relationships among these entities. This deficiency underscores a significant disparity between machine intelligence and human-level cognition. Primarily, it stems from the absence of key concepts such as intuitive understanding of physical phenomena and the causative links between them. Consequently, bridging this gap necessitates a paradigm shift towards integrating causal reasoning frameworks into AI and ML methodologies.

4. Compositionality:

In this project, compositionality will serve as a pivotal implementation method. Typically utilized in algorithms such as one-shot learning and few-shot learning. Compositionality stands as a technique enabling models to learn from sparse data (Lake et al., 2017). This approach is closely intertwined with *learning-to-learn*, also known as *transfer learning* or *meta-learning* (Lake et al., 2011). Notably, compositionality and learning-to-learn are intricately interconnected concepts.

The essence of compositionality lies in the notion that learning from individual components of a system can expedite the learning of a different system with similar components as a whole, particularly in scenarios where data availability is limited. By initially acquiring knowledge of constituent elements, subsequent learning endeavors can capitalize on this foundation. Thereby, this can facilitate more efficient learning processes with minimal data input. Indeed, the ability to learn from sparse datasets represents a paramount challenge in the pursuit of developing machines with human-like cognitive capacities. As such, *compositionality* emerges as a crucial tool in the quest to bridge this disparity between artificial and human intelligence.

5. Learning-to-learn:

Learning-to-learn, also referred to as *transfer learning* or *meta-learning*, encompasses a nuanced distinction depending on the context in which it is employed. Within the realm of Deep Learning, transfer learning or meta-learning typically involves the utilization of pre-trained layers to expedite the learning process for subsequent tasks. However, in the domain of Cognitive Science, learning-to-learn embodies a more profound connotation aligned with its literal interpretation.

Consider a child's acquisition of knowledge regarding the composition of letters or digits, recognizing them as distinct arrangements of discrete shapes. Beyond mere task-specific application, this cognitive process engenders a fundamental understanding that entities can be comprised of constituent components, thereby facilitating discrimination and comprehension. This form of learning transcends *pattern recognition*, engendering the assimilation of *novel concepts* and

the construction of *cognitive models*. Consequently, individuals develop the capacity to learn *how to learn*, a trait embodied by their adeptness at deriving insights from sparse data.

The efficacy of learning-to-learn lies in its facilitation of not only task-specific adaptation but also the acquisition of generalizable cognitive frameworks. By embracing the underlying principles of compositional learning, individuals become adept at extrapolating knowledge across diverse domains, thereby enhancing their capacity for rapid and efficient learning. Thus, learning-to-learn emerges as a cornerstone in the endeavor to equip artificial systems with human-like cognitive capabilities, enabling them to thrive in dynamic and data-scarce environments.

What's Novel:

First of all, a comprehensive review of prior research endeavors within the field will be conducted to elucidate the existing landscape of knowledge. This critical examination will encompass an exploration of how foundational concepts such as *causality*, *compositionality*, and *learning-to-learn* have been leveraged in the context of one-shot and few-shot learning paradigms. The rationale behind this inquiry lies in the recognition of these cognitive principles as potential differentiators between human and machine learning processes, with a particular emphasis on model building in comparison with pattern recognition. Consequently, the endeavor to learn from sparse data emerges as a natural consequence of portraying the differences between human and machine learning modalities. Thereby, it serves as a potential indicator of progress towards achieving human-level intelligence.

The novel aspect of this project, unlike former studies, lies in its innovative approach to classifying handwritten characters using machine learning techniques, specifically leveraging the principle of *compositionality*. While previous research, such as the work by Lake et al., (2017), has demonstrated the efficacy of Bayesian Program Learning (BPL) in achieving human-level performance in one-shot classification tasks, this project deviates from using BPL. Instead, its aim is to harness the concept of *compositionality* by both employing different data processing techniques and using Machine Learning models, instead of BPL. To capture causality, Lake et al., (2015) used Bayesian Program Learning (BPL). However, with this project, we can hope to capture causality by employing chronological order of strokes in two dimensional space in the Omniglot data.

By refraining from reliance on statistics or conditional probability which are inherent in BPL, the project intends a quest to conceptualize and exploit *compositionality* and *causality* as the cornerstone of its algorithmic framework. This departure necessitates a creative reimagining of how compositionality and causality can be translated into a practical algorithmic implementation. Unlike BPL, which utilizes a causal and compositional model inherently, the proposed algorithm of this project seeks to extract the essence of compositionality and causality; and embed them within the learning process without direct recourse to probabilistic methodologies.

Lake et al., (2011) created a library of strokes. They clustered 40,000 strokes by using k-means to form 1000 centroids. The library is composed of these 1000 centroids. Then, they tested three different models: the stroke model which in fact is a generative model that they introduced, the Deep Boltzman Machine (DBM), and Nearest Neighbor (NN) for 20 way-classification from one example. The stroke model (% 54.9) outperformed both (% 39.6 for the DBM and % 15.7 for the NN).

Li et al., (2022) reconstructed Omniglot data into videos. By employing spatial and temporal information in txt files, they reconstructed the text record of strokes as a video of writing tracks. I am going to use a similar technique. They also employed linear interpolation algorithm to complete the data in milliseconds to reconstruct the character writing as accurate as possible. However, since I am not going to use the time dimension in my model so as to require any interpolation method. I am only going to use spatial knowledge in txt files to construct *components* of images and use time dimension only to put these newly created images into a chronological order.

Thus, the novelty of this endeavor lies in its ambition to forge new pathways in AI and ML by leveraging the inherent principles of *compositionality* and *causality* to achieve superior performance in classifying and even predicting handwritten characters from sparse data by employing a novel data processing and implementation approach. This departure from traditional statistical approaches represents a paradigm shift towards more conceptually driven algorithmic solutions, potentially unlocking novel avenues for advancing the capabilities of machine learning systems.

Implementation of the Project:

1. The Dataset :

Each process of implementation of the project is going to be clarified as step by step. Specifically, a Machine Learning model will be trained to discern and learn components (subparts) of a dataset called Omniglot which composed of 1623 distinct handwritten characters from 50 different alphabets. The Omniglot dataset aims to cultivate algorithms that mimic human learning processes more closely. Each character was independently drawn by 20 individuals through Amazon's Mechanical Turk platform. Each image is paired with stroke data, presenting sequences of [x,y,t] coordinates where time (t) is measured in milliseconds.

Important Note : The dataset is used mostly in generative models like BPL (Lake et al., 2011, 2015 and 2017), and in those models, it requires parsers. In other models, it is generally used for its spatial and temporal aspect (Li et al., 2022). I invented a new method which seems easier to understand and apply to.

2. Pre-processing :

Each picture is 105 by 105 pixels. Each pixel has a binary value : zero or one. Therefore, there is no requirement for the preprocessing for these images. However, stroke data composed of txt files. These are comparatively hard to process. There are many different ways to implement these txt files. For example, they can be used in generative models like Bayesian Programing Learning (Lake et al., 2015, and 2017) , or in studies like spatio-temporal sparse few-shot learning⁵. However, this project, for two reasons, refrain from both kind of approach. First, this project will use Machine Learning techniques instead of Bayesian Programing Learning . Second, this project's aim is to rigidly stay in the domain of Cognitive Science, therefore, second approach is also not going to be used. To this end, I created a new method to process this data. Therefore, this project is novel both in terms of data processing techniques and implementation. This will soon become more clear.

3. Txt Files :

These files start with the word “START” and they have three columns at each row. First two columns for trajectories of strokes in the x-y coordinate system, and third column is for the time axis. These values, number of rows, continues until strokes finish. When a stroke ends, you encounter with the word “BREAK”. That can be understood as human drawers finished one stroke and will start to another. Then rows continue until another BREAK. All txt files start with the word “START” and end with the word “BREAK”.

```
START
11.745075,-15.097315,0
13.677961,-14.130872,139
14.644404,-14.130872,147
.
.
.
45.66777, -34.43333, 345
BREAK
.
.
.
BREAK
```

This is first stroke.

Each numeric value in the rows for the first two columns (spatial values) can only range from -105 to +105. However, that is not usually the case. Because strokes are generally at the center of the pictures.

Each stroke will be turned into a png image of 105 by 105 pixels. Then, each stroke image will have one channel, and after turned into arrays, their values also will be turned into values between 0 and

1. Therefore, in the end, input will be an image and output will be many images that are different strokes (components) of the input image.

For example : This is the first character from 'Early_Aramaic' alphabet.



When it is plotted with stroke data :



As you can see, colors are different. Because there is an order in drawing of strokes as mentioned before, when *txt* files are explained. Stroke order determines which color to use : first stroke is red, second stroke is green, third stroke is blue, etc. This order can be used to capture “*causality*”. However, since this project will process data in a different way, we hope to capture causality in a different way, mainly by taking advantage of the order of pictures that are going to be created.

I am going to create pictures of these components to capture the *compositionality* and *causality* by using data in *txt* files. For example, I created components below by employing *txt* files.



As you can see, they are black. Because we do not need to RGB channels. We need only one channel. And these images are 105 by 105 pixels exactly like input images. Because creating pictures with a different size may change the scale of the location of the strokes with respect to the whole area. I wanted to be sure whole pictures overlap on each other to save the scale.

4. Network Architecture:

If I manage to use variable-sized outputs, I am going to (start to) use Scikit-Learn machine learning models. However, if I would not manage to use variable-sized outputs, I am going to fix the length. The pictures I would create in the end will be merged into one big array. Length will be increased by filling arrays with zeros until they reach to “the max length of strokes * 105 * 105”. For example, if maximum length of strokes, the picture with maximum length of strokes, or maximum number of strokes between each BREAK from START to the last BREAK in txt files, is 5. Then, the length of outputs will be $5 * 105 * 105$. In this way, the model will learn which characters are composed of which components.

In case, even if I would fix the length and still would not manage to train a model with scikit-learn or I would manage and still have more time to develop the project. I am going to implement more advanced models by using PyTorch. More specifically, I am going to use a Convolutional Neural Network (CNN) which are commonly used for image-related tasks due to their ability to effectively learn hierarchical features from input data. I am planning to use an encoder-decoder architecture, probably U-Net or SegNet, which is commonly used for image segmentation tasks. The encoder part of the network will downsample the input image to capture high-level features, while the decoder part will upsample the features to generate the segmented output images.

Note : As mentioned earlier, Omniglot dataset is composed of 1623 distinct handwritten characters from 50 different alphabets. Each character has 20 different png images. However, after our augmentation of images, number of pictures will be increased.

5. Training and One-shot classification:

The model will be trained on 30 of those alphabets. After training, we hope it will be to have learned which image is composed of which parts. Then, this pre-trained model will be trained only with one instance (sparse data) from each one of the remaining 20 alphabets (validation data) that it did not see before. Then it will be used for one-shot classification by comparing the outputs of each image with a distance metric. This is called 20-Way one-shot classification which is a type of N-way classifications. In the end of this classification, there will be one image that will be the most similar to the validation image compared to the rest of the (N-1) pair of images.

To summarize, train and test set of one-shot classification will correspond to each other one by one. For example, it will be trained for a character of a language's alphabet (let's say 'x' for Latin alphabet) which it was not trained on. It was neither trained on 'x' nor any other character of Latin before. Because characters in alphabets have similar structures. Therefore, our choice makes one-shot classification task harder. Then, it will be trained for other 19 characters from other 19 alphabets, one from each. Then, we will expect it to classify the correct character which it saw just once. It will also classify other 19 wrong characters and we will assign a distance metric to this classification. Subsequently, results will be compared to other classification models' outputs. This

comparative analysis aims to elucidate the efficacy of *compositionality* and *causality* as a mechanism for enhancing learning performance in resource-constrained environments.

In essence, the implementation strategy of the project is structured to iteratively explore and leverage foundational cognitive principles, culminating in the operationalization of *compositionality* and *causality* as a pivotal mechanism for fostering enhanced machine learning capabilities. Through this systematic approach, the project endeavors to contribute towards the advancement of AI systems towards achieving human-like cognitive proficiencies.

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References :

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